A Comprehensive Survey on Federated Learning and Decentralised Access Coordination for Machine Learning

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***Abstract*—Federated Learning (FL) has become a core strategy for training machine-learning models over sensitive and continuously expanding data without centralizing raw records. This survey consolidates current (2022–2025) technical directions across horizontal, vertical, and transfer-based FL, with emphasis on blockchain-backed coordination and DAO-style governance. We outline optimization fundamentals (e.g., FedAvg), performance tradeoffs, and deployment characteristics in IoT, finance, healthcare, and industrial systems driven by high-volume, high-velocity, and non-IID data. Continuous data growth, inter-institution dependencies, and privacy regulation have accelerated adoption of secure aggregation, differential privacy, and cryptographic protocols. The review highlights how decentralized mechanisms enhance auditability, fairness, and trust, while introducing new operational burdens such as communication overhead, bias propagation, participation imbalance, and legal constraints. We further examine access coordination models using blockchain and DAO governance, where smart-contract-based control enables transparent model update logging, incentive mechanisms, and distributed decision-making. The work concludes that scalable privacy guarantees, policy-aligned data governance, and incentive-aligned coordination are pivotal for next-generation FL ecosystems across high-risk and data-dense environments.**

***Index Terms*—federated learning, decentralised coordination, blockchain, DAO, IoT, privacy-preserving ML**

I. Introduction

Federated learning is a decentralized machine learning paradigm that enables collaborative model training across multiple entities without sharing raw data, effectively enhancing privacy, data governance, and fairness, especially when integrated with blockchain-based data pipelines in crowd sourced environments.

A. Origin and back story

The concept of federated learning originated in response to increasing privacy regulations (like GDPR, CCPA) and practical needs in distributed, data-rich domains such as IoT, healthcare, and mobile networks. Early machine learning centralized all data, raising regulatory and privacy issues. The federated approach began gaining traction around 2016, notably with Google's initial work enabling collaborative neural network training directly on users' devices without uploading personal data. Over time, researchers built on this decentralized scheme to ensure not just privacy, but also improved fairness and trustworthiness in collaborative AI.

II. Technical explanation

Federated learning fundamentally involves multiple clients, each with their own local data. Let $D = \{D_1, D_2, \dots, D_n\}$ be local datasets for n clients. Each client trains a local model on D_i, and periodically communicates model updates (not data) to a central server.

The canonical optimization objective is:

$$\min_w F(w) = \sum_{i=1}^{n} \frac{|D_i|}{|D|} F_i(w). \tag{1}$$

A common update step (FedAvg algorithm) for client i with learning rate η and local step t is:

$$w_i^{t+1} = w_i^t - \eta \nabla F_i(w_i^t). \tag{2}$$

Aggregation at the server is typically:

$$w^{t+1} = \sum_{i=1}^{n} \frac{|D_i|}{|D|} w_i^{t+1}. \tag{3}$$

III. Types of Federated Learning (As of 2025)

A. Horizontal Federated Learning (Sample-based)

Clients have datasets with similar feature spaces but different samples (users). Evolved from traditional settings like hospitals sharing patient records with the same variables but different patients. Focuses on aggregating models trained independently on similar data structures. Example: Multiple banks collaboratively build a fraud detector using locally collected transaction data.

B. Vertical Federated Learning (Feature-based)

Clients have different feature spaces but share users (samples). Became crucial with businesses (e.g., a retailer and a bank) collaborating where user overlap exists but data types differ. Secure feature alignment, homomorphic encryption often used for privacy. Example: A supermarket and a credit card company jointly build a consumer behavior model.

C. Federated Transfer Learning

Clients differ in both samples and features. Uses transfer learning to adapt global models to new local data. Example: A small regional clinic joins a federated healthcare network with different data and a small patient base.

D. Blockchain-Enhanced Federated Learning

Emerged to tackle trust, fairness, and accountability via immutable ledgers. Smart contracts record model updates and 'unlearning' actions, maintaining auditability and fairness in crowdsourcing pipelines.

IV. Performance and Use Case Comparison

TABLE I
PERFORMANCE AND USE CASE COMPARISON (SUMMARY)

Method	Privacy	Use Case	Trust/Audit
Horizontal FL	Strong	Healthcare	Moderate
Vertical FL	Strong	Finance/Retail	High
Transfer FL	Adaptive	Small org	Moderate
Blockchain-Enhanced	Very Strong	IoT/Crowdsourcing	Very High

V. Continuous And Ever-Growing Data

Continuous and ever-growing data: It refers to the information that is being created nonstop and keeps expanding over time. Unlike static datasets (where data is collected once and it doesn't change much), this kind of data is always being updated (every minute, hour, or day) from many sources. It grows larger and more complex as new inputs arrive. The data is continuously generated from million or billions of smart devices like healthcare, finance, wearables, all generating information constantly, making it high volume, high velocity and highly diverse .GPS location, or medical readings, this data is flowing in every second. It's huge in scale, coming from all around the world, and it's very diverse because each person's habits and environment are different. Instead, each device learns locally and only shares the "learning," keeping your personal data private [1]. Think about all the purchases people make transaction at restaurants, grocery stores, travel sites. Every transaction is recorded, and over time, this becomes an enormous collection of data points. Millions of customers and tens of thousands of merchants produce transaction records every day, and year long. These records reveal patterns like when people shop, how much they spend, or where they go during holidays [2]. You've probably seen how your credit score changes over time like loans, repayments, new accounts, inquiries. Multiply that by millions of people and multiple banks, and you get an ever-expanding pool of financial information. Instead of sending all that sensitive data to one place, it explores how different banks can securely combine their information, learn together, and calculate credit scores while keeping customer data private [3]. Companies like banks and online shops can work together without sharing their customers' sensitive data. Normally, this kind of collaboration would be risky by exposing payment details or personal information. But using advanced encryption, it's possible to analyze data while keeping it locked up. Even when billions of data points are involved, it's possible to spot fraud or gain insights without ever opening the "data vault." Big Data here means huge amounts of encrypted, continuously updated information that needs smart and secure processing [4]. Let's take a social network like Facebook or a recommendation system like Amazon new users, new products, and new interactions are added every day. The connections form complex webs that grow larger and more complicated over time. Labeling data (like identifying interests or patterns) can be expensive and slow. This paper shows how you can pick the most useful pieces of information from these large networks to learn effectively without needing to process or label everything. It's Big Data because these networks are huge and constantly expanding [5]. Industrial systems like smart factories, and energy grids generate continuous streams of sensor data, with high volume, velocity, and variety. Factories don't stop working machines run around the clock, sensors monitor temperature, pressure, or vibrations, and systems adjust in real time. The paper talks about how companies can use this information to predict when a machine might fail or how to optimize production without sending all the data to a central server [6]. [1].

VI. Sources of Continuous and Ever-Growing Data

A. 1. From Smart Devices like Phones, Watches, and Health Trackers

Source: Smartphones, wearable fitness bands, medical devices, and other gadgets that people use daily. Whenever you walk around with your phone in your pocket, or you track your heartbeat with your smartwatch, data is being recorded. These devices monitor your location, movement, sleep, and other health metrics. Every person using these devices adds new information every second, and with billions of users, it becomes a massive source of data. Because this data is personal, companies must protect it carefully.

B. 2. From Financial Transactions like Credit Cards and Digital Payments

Source: Banks, online stores, shopping malls, restaurants, and travel services. Whenever you make a purchase, swipe your card, or pay online, details such as the amount, time, and place are logged. Multiply that by millions of customers and thousands of stores — that's a vast amount of data. Each transaction contributes to understanding customer habits, spotting fraud, and improving services.

C. 3. From Customer Records and Credit Reports at Banks

Source: Financial institutions such as banks, loan providers, and credit agencies. Opening a bank account, taking a loan, or making payments all add to an individual's financial profile. This data isn't static — it grows with each transaction. Banks combine this information to assess creditworthiness, ensuring that privacy and security are maintained.

D. 4. From Encrypted Customer Interactions in Finance and E-commerce

Source: Payment platforms, online stores, banks, and insurance companies. Beyond purchases, companies track browsing behavior, payment methods, and login patterns to detect suspicious activities. This continuously evolving data is protected using encryption, enabling secure analysis without compromising user privacy.

E. 5. From Networks like Social Media and Recommendation Systems

Source: Social networking platforms, video streaming services, and online marketplaces. Every post, video click, or product purchase adds new data. As more users join daily, new relationships and patterns emerge. This expanding graph of interactions helps platforms personalize content and recommendations, though labeling and managing such data is computationally intensive.

F. 6. From Industrial Sensors in Factories and Machines

Source: Manufacturing plants, energy grids, and smart supply chains. Sensors in industrial systems record parameters like temperature, pressure, and vibration in real time. These readings accumulate rapidly, and local edge processing is often used to handle them efficiently. Since factories run 24/7, this data grows continuously, making decentralized processing essential.

VII. Characteristics and Examples

A. Federated Learning with IoT Devices

High volume – Millions or Billions of devices create massive amounts of data daily. Distributed – Data is scattered across devices like smartphones, wearables, and medical sensors. Non-IID – Each user's data is different depending on their location, lifestyle, or habits. Privacy-sensitive – The data often includes personal information like health metrics or location. Time-dependent – Data is continuously updated and depends on events happening in real time.

B. Credit Card Transactions

Temporal patterns – Spending habits show daily, weekly, and seasonal trends. Large scale – Millions of transactions occur across thousands of merchants. Dynamic – Consumer behavior changes over time with new purchases and trends. Financial sensitivity – The data includes sensitive payment details and spending patterns.

C. Industrial IoT and Smart Factories

High volume and velocity – Machines and sensors produce continuous streams of data without pause. Multi-sensor data – Different types of sensors create diverse datasets. Operational relevance – Real-time monitoring is needed to prevent breakdowns and optimize performance.

VIII. How It Will Be Handled

Apache Spark: Is one of the most popular tools used to handle large amounts of data, especially when that data is continuously growing and needs to be processed quickly. What Apache Spark Does: Processes Big Data super fast, Handles streaming data in real time, Works with many data sources, Offers powerful tools for machine learning and analysis.

Besides Apache Spark, several other tools and technologies are widely used to process, store, and analyze large datasets that especially that are always growing or arriving in real time. Below is a list of these tools:

- Apache Hadoop
- Apache Kafka
- Apache Flink
- NoSQL Databases (MongoDB, Cassandra, Redis)
- Google BigQuery / Amazon Redshift / Snowflake
- TensorFlow / PyTorch

IX. Challenges of Handling Continuous and Ever-Growing Data

- Volume – The Data is Just Too Big
- Velocity – It Arrives Too Fast
- Variety – Different Types of Data
- Veracity – The Data Can Be Messy or Inaccurate
- Scalability – Growing Data Needs Growing Systems
- Integration – Bringing Everything Together
- Real-Time Decision Making – Acting Fast Enough

X. Benefits and Tradeoffs

Benefits include smarter decisions with real-time insights, personalized experiences, better efficiency, enhanced customer satisfaction, stronger security and fraud prevention, innovation and new opportunities, compliance and trust. Tradeoffs include Speed vs. Accuracy, Privacy vs. Data Usefulness, Cost vs. Scalability, Real-Time Processing vs. Resource Use, Centralization vs. Distribution, Model Complexity vs. Explainability.

XI. Interdependent Data Among Different Stakeholders

In today's connected world, data no longer sits within the walls of a single organization. Banks, retailers, manufacturers, and even regulators actively exchange information that crosses industries and borders. This constant sharing creates what researchers call *interdependent data* — information that is shared, correlated, and mutually influential across stakeholders. Unlike isolated datasets of the past, interdependent data captures relationships that shape real decisions in finance, retail, fraud detection, and industrial IoT. This paper draws on six important studies [1], [3], [4], [6], [12], [13] to explore these interdependencies, showing how federated learning, secure machine learning, and graph-based modeling help organizations unlock value while managing challenges such as bias, privacy, and regulatory uncertainty.

A. What Is Interdependent Data?

In today's digital ecosystems, data does not exist in isolation. Stakeholders such as banks, manufacturers, and retailers rely on shared datasets that influence one another. This interdependent data drives decision-making, improves predictions, and enhances systemic resilience. Studies show that federated learning [4], [12], collaborative machine learning [6], and graph-based active learning [13] represent new paradigms for handling such data.

Interdependent data consists of information that multiple stakeholders share, correlate, and rely upon. It creates mutual dependencies where one institution's records affect decisions across the ecosystem. For example, repayment data shared across banks builds accurate credit profiles [1]. Federated learning enables organizations to analyze such data collaboratively without centralizing raw inputs [4]. In industrial IoT, connected sensors across suppliers, manufacturers, and distributors create real-time dependencies that optimize production schedules and demand forecasting [12].

B. Real-World Examples

Researchers demonstrate interdependent data in multiple sectors. In finance, credit bureaus aggregate repayment data, where a single missed payment reported by one bank lowers trustworthiness across the system [1]. In fraud detection, Visa's collaborative machine learning integrates merchant, customer, and bank data without exposing raw records, producing holistic fraud profiles [6]. In retail, merchant category analysis reveals that customer transactions in one domain, such as restaurants, strongly correlate with purchases in another, like cinemas [3].

In industrial IoT, interdependent data flows across logistics and manufacturing nodes, and supply-chain machine failures propagate delays through dependencies:

$$P(\text{delay}) = 1 - \prod_i (1 - p_i), \tag{4}$$

where p_i represents the failure probability of each node i. This cumulative function models ripple effects in interdependent systems [12].

Interdependent data also drives innovation in healthcare-like collaborative environments, which are conceptually similar to federated learning systems. Research on federated learning [4], [12] shows that hospitals and healthcare providers can train predictive models for diagnosis without pooling raw patient data. Just as banks collaborate on credit risk [1], hospitals share encrypted model updates that contribute to a stronger global model. This flow of interdependent data ensures better treatment outcomes and preserves privacy, reflecting the same principles outlined in industrial IoT and financial systems.

C. Smart Infrastructures and Urban Systems

Smart infrastructures provide another strong case of interdependent data. Studies on federated learning for IoT [12] and graph-based active learning [13] emphasize how connected devices and nodes constantly depend on one another. In transportation or smart-city contexts, sensors at intersections feed into traffic optimization models, while ride-sharing platforms and logistics providers update their systems in response. Similar to how merchant categories reveal hidden spending correlations in retail [3], these interdependencies in urban systems allow real-time coordination. By applying graph neural networks [13], such networks capture relationships between distributed nodes, ensuring collective decision-making that improves efficiency across the ecosystem.

D. Usefulness in Data Science

Data science benefits significantly from interdependent data. Federated learning enhances collaborative intelligence by combining distributed data into global models without breaching privacy [4], [12]. Privacy-preserving techniques such as secure multiparty computation and encrypted model aggregation ensure institutions contribute data safely [6]. Graph neural networks capture complex dependencies in relational datasets, improving predictive performance [13].

By pooling interdependent datasets, organizations increase accuracy, identify hidden correlations, and drive systemic innovation across sectors. Interdependent data actively supports real-time decision-making in fast-changing environments. In industrial IoT networks, for example, sensors placed across supply chains constantly stream information that stakeholders analyze together. Using federated learning, organizations generate predictive insights about equipment failures or sudden shifts in demand without exposing their raw operational data [12]. This collective approach allows manufacturers, suppliers, and distributors to benefit from shared intelligence while still protecting competitive confidentiality.

Because interdependent data flows capture the broader system context, stakeholders can respond to disruptions much faster than they could with isolated datasets. Interdependent data also strengthens the robustness and fairness of AI models. When multiple organizations contribute diverse datasets, models avoid overfitting to narrow patterns or inheriting a single institution's bias [4]. In financial credit scoring, for instance, pooling repayment data across banks [1] leads to more accurate and fair evaluations of borrowers. Likewise, in fraud detection [6], analyzing transaction signals across merchants uncovers anomalies that would appear normal in a single dataset. By drawing from shared but distributed sources, interdependent data ensures that insights are not only more accurate but also fairer and more widely applicable across populations.

XII. Simulation Example

A simulation of federated credit scoring demonstrates the value of interdependent data. Banks train local models on repayment data and send anonymized updates w_i to a federal aggregator. The global model is defined as:

$$W = \sum_i w_i. \tag{5}$$

XIII. Identifying and creating correlation

Mutual information, dependency graphs, and covariance matrices can quantify relationships. For two stakeholders with datasets X and Y, the Pearson correlation coefficient is:

$$\rho(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y}. \tag{6}$$

XIV. Drawback and Negative Side

Interdependent data systems face multiple drawbacks. Bias in one dataset can propagate across the network, reinforcing inequalities [4]. Collaborative systems demand high computational and communication costs [12]. Privacy-preserving computation, while promising, remains resource-intensive and vulnerable to adversarial strategies [6]. Legal uncertainties surrounding cross-institution data sharing further complicate adoption. Operational dependencies also create risks: one node's disruption in an IoT network cascades across the ecosystem, impacting production and customer satisfaction [12].

Another drawback is that interdependent data can unintentionally create unfair power dynamics among stakeholders. For example, in federated learning setups, larger organizations contribute bigger datasets and therefore dominate the global model [4]. Smaller players may feel excluded, even though their niche data adds unique value. This imbalance reduces trust and discourages collaboration, limiting the true potential of interdependent systems.

Interdependent data systems also slow down responsiveness because of their reliance on secure protocols. Visa's work on collaborative machine learning shows that encrypted joins and secure multiparty computation add noticeable overhead [6]. While these methods preserve privacy, they delay decision-making in time-sensitive domains like fraud detection or industrial IoT. Stakeholders face a tradeoff between faster actions and stronger privacy safeguards, making adoption more difficult.

Finally, interdependent data raises challenges in transparency and accountability. In Industrial IoT (IIoT) supply chains, when a delay occurs, it is often unclear which stakeholder's dataset or decision triggered the disruption [12]. Similarly, in financial ecosystems, shared scoring models [1] make it difficult to assign responsibility if errors or biases appear. Without clear accountability mechanisms, trust in these systems weakens, and organizations hesitate to share sensitive data openly.

XV. DAO CONCEPT

Decentralized Autonomous Organizations (DAOs) are changing how groups make decisions by replacing leaders and managers with collective decision-making. Instead of depending on executives or boards, DAOs let members vote and act together using clear rules written in smart contracts. DAOs started in the blockchain world, but today their ideas go far beyond cryptocurrency. We can already see similar approaches in federated learning, where organizations safely share insights without giving away raw data.

XVI. Introduction to DAO Governance

DAOs take the federated idea further by applying it to governance. A DAO governance decision can be modeled as:

$$D = f(V, T, R), \tag{7}$$

where D is decision, V votes, T token/reputation weights, and R contract rules.

XVII. Concept & Working

Analogies: A DAO works like a co-op supermarket. Members vote on which products to stock; rules (smart contracts) define how votes are counted and decisions are made.

XVIII. DAO Types in 2025

By 2025, DAOs include: Protocol DAOs, Investment DAOs, Service DAOs, and Social DAOs. These mirror federated and collaborative systems and coordinate resources through token-weighted governance and automated proposals.

XIX. Demonstration with Examples

Finance DAO, Fraud DAO, and Research DAO examples illustrate how members vote on lending policies, fraud detection rules, or project funding while protecting data privacy.

DAO evolution can be described using Markov chain models; the next state depends on current state and transition probabilities.

XX. Information and Insight Control

Reputation update model:

$$R_i^{t+1} = R_i^t + \alpha C_i - \beta D_i, \tag{8}$$

where C_i constructive contributions and D_i dishonest actions; α, β are scaling constants.

XXI. Challenges

DAOs face many of the same challenges that exist in other distributed systems. As more members join, scalability becomes a serious issue — communication takes longer, decisions slow down, and operational costs rise [4], [12]. DAOs also remain vulnerable to Sybil attacks, where a single actor creates multiple fake identities to manipulate outcomes and undermine fairness [6].

A third major challenge is coordination failure. When members do not participate actively, the system cannot function effectively, even if the governance rules are strong [4]. This issue parallels non-participation in federated learning, where inactive clients reduce global model performance.

We can capture this problem with a simple game-theoretic model:

$$U_i = pB - c, \tag{9}$$

where U_i represents the utility (or benefit) for an individual member, B is the shared collective benefit, p is the probability that other members will participate, and c is the cost of joining or contributing effort.

If $U_i < 0$, rational members will stop contributing — similar to how people drop out of group tasks when participation seems low. In practical terms, it's like planning a group trip: if too few friends commit, even the enthusiastic members lose motivation, and the effort collapses. This dynamic highlights the fragility of decentralized cooperation in DAOs and federated governance systems.

XXII. Security, Privacy and Fairness

Threats include poisoning attacks, membership inference, model inversion, and adversarial manipulation. Defenses include differential privacy, secure aggregation, robust aggregation, and cryptographic protocols.

XXIII. System Architecture and Tools

Popular stacks include Apache Spark, Kafka, Flink, NoSQL, BigQuery/Redshift, TensorFlow/PyTorch. Edge processing, pruning, and model compression mitigate communication costs.

XXIV. Conclusion

Interdependent data forms the backbone of digital collaboration across sectors. Federated learning and DAO-like governance offer pathways to privacy-preserving, auditable, and collaborative AI. Future work should address fairness, scalable privacy, legal harmonization, and efficient incentive design.

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